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A multi-scale wavelet and spectral kurtosis-guided framework for noise-resistant rolling bearing fault diagnosis

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ABSTRACT

The health status of rolling bearings is essential for ensuring the safe and stable operation of industrial equipment. However, vibration signals acquired under actual operating conditions are often contaminated by strong background noise. This contamination makes it difficult for traditional fault-diagnosis models to effectively extract subtle fault features, leading to low diagnostic accuracy and poor noise robustness. To address this issue, this paper proposes a novel noise-resistant diagnostic network based on multi-scale wavelet convolution and kurtosis guidance (MWKG-Net). It is the first network to employ the lightweight Multi-Scale Wavelet Convolution Block (MWConv1D) to adaptively decompose one-dimensional vibration signals into multiple frequency subbands. This design not only enlarges the receptive field and achieves preliminary noise suppression, but also captures multi-scale fault information precisely, while reducing model parameters and computational overhead. Furthermore, we design a Kurtosis-Guided Feature Re-weighting (KGFR) module that quantifies the prominence of fault impulses by computing the spectral kurtosis of feature maps and dynamically re-weights feature channels. This enhances fault-related impulse components and further suppresses noise and unrelated harmonics. By combining wavelet-based multi-resolution analysis with kurtosis-based impulse sensitivity, the proposed MWKG-Net significantly improves the extraction of robust fault features under strong noise. Moreover, the model's lightweight design makes it suitable for deployment on resource-constrained industrial platforms. Experimental results on the MFS-MG, BJTU-RAO and Urban Rail Train datasets demonstrate that, compared with state-of-the-art methods, our model maintains very high diagnostic accuracy even under extremely low signal-to-noise-ratio conditions, showcasing exceptional noise robustness, computational efficiency, and generalization capability.

1. Introduction

Rotating machinery is indispensable power-transmission equipment in modern industrial systems [1,2], where its stable operation

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and effective fault diagnosis are critical for ensuring production efficiency and safety [3–5]. As core support components, the health of rolling bearings directly affects the performance and lifespan of the entire system [6,7]. Bearings, however, often operate under harsh and variable conditions such as high temperature, humidity, and heavy loads. These factors can lead to various fault modes, including pitting, spalling, and cracking. Vibration signals collected by sensors provide the most direct basis for assessing bearing condition [8]. Yet, these signals are typically nonlinear, non-stationary, and unavoidably contaminated by strong background noise from other components or the environment. This noise poses a significant challenge by severely obscuring faint early fault features, making accurate and reliable fault diagnosis difficult.

Early-stage fault diagnosis predominantly depended on conventional signal-processing techniques such as Fourier analysis, wavelet analysis, and empirical mode decomposition [9,10]. Under certain conditions, these approaches can achieve satisfactory diagnostic performance. For example, YANG et al. [11] developed a weighted bandwidth decomposition method (WBD) based on the concept of spectral segmentation and reconstruction, Gámiz et al. [12] proposed a double Markov chain–based approach for system health-state prediction, leveraging real-time sensor data to provide early warnings of potential failures. Xin et al. [13] proposed the equal-proportional expansion optimization-gram (EPEOgram) method, reconstructing spectral coherence to automatically identify cyclic frequencies, locating center frequency bands, and generating ODBs clusters for bearing fault detection, enhancing multi-frequency band identification without prior fault information. Xin et al. [14] proposed a parameter-adaptive feature mode decomposition (PAFMD) method, using spectral trend band division to initialize FIR filters with ASNR for period estimation, and a coarse-to-fine filter length strategy to optimize fault feature extraction, enhancing multi-fault information robustness and noise immunity. Xu et al. [15] proposed a maximum L-Kurtosis deconvolution (MLKD) algorithm, leveraging L-Kurtosis invariance with an iterative filter and frequency-domain filtering (FDF) to reduce noise, with experiments demonstrating improved noise immunity and robustness. However, these traditional methods have significant drawbacks. They typically require extensive expert knowledge and signal-processing experience to manually design feature extractors. This process is time-consuming, labor-intensive, and difficult to automate for rapid, intelligent fault diagnosis.

In recent years, propelled by rapid advances in artificial intelligence, deep learning–centric approaches have consistently delivered breakthroughs in areas like image recognition and natural language processing. Many researchers have also begun to introduce deep learning into the field of fault diagnosis [16–18], designing numerous models based on convolutional neural networks (CNNs), recurrent neural networks (RNNs), and the like, thereby realizing end-to-end automatic feature extraction and fault classification directly from raw signals and markedly improving diagnostic efficiency and accuracy. Du et al. [19] proposed a frequency mask and decoupling max-logit-based XAI method (FM-Explainer), optimizing an input frequency-domain mask to identify fault features and introducing decoupling max-logit regularization with a spatial penalty to avoid unreliable OoD explanations; their explanations align closely with characteristic fault frequencies. Zhou et al. [20] proposed an end-to-end fault-diagnosis framework (DRSwin-ST) based on a sparse transformer, combining a multi-scale architecture to extract fault features and improve fault-identification accuracy. Guan et al. [21] proposed an intelligent fault diagnosis method based on information fusion and a parallel lightweight convolutional network, employing a normalized pulse energy kurtosis weighted rule for multi-channel signal fusion and a novel lightweight CNN for feature extraction and classification, thereby improving diagnostic accuracy and precision. Wang et al. [22] proposed an improved spiking neural network (ISNN) for intershaft bearing fault diagnosis, encoding raw vibration data into spike sequences, deriving and replacing invalid gradients, compensating for forward propagation information loss and simplifying backpropagation; this ISNN outperformed previous SNN methods and conventional second-generation neural networks. Mehta et al. [23] proposed a federated learning–based duplet classifier framework that decomposes mixed fault classification into parallel component-specific networks; this method achieved excellent mixed fault classification accuracy under highly unbalanced and heterogeneous fault label distributions. Li et al. [24] proposed a Multilayer Grad-CAM (MLG-CAM) method that leverages gradient flows from multiple convolutional layers to generate multi-resolution activation maps, highlighting *cyclo*-stationary impulses in the time domain and emphasizing fault characteristic frequencies in the frequency domain, thereby enhancing neural network explainability.

Despite these advances, most deep learning models suffer a dramatic performance drop when confronted with strong noise interference. This vulnerability arises because standard convolution operations, while adept at extracting local structural features, are sensitive to noise and prone to misclassify noise as valid features. To address this, researchers have developed more complex, noise-robust models. Wang et al. [25] proposed an innovative Neural-Transformer that integrates a frequency-slice wavelet transform (FSWT), a separable multiscale spiking tokenizer (SMST), and multi-head spatiotemporal spiking self-attention (MHSSSA) to achieve high-precision, robust fault diagnosis at low computational cost. Han et al. [26] introduced a Transformer augmented with channel-attention mechanisms, achieving strong diagnostic accuracy under high-noise conditions; however, their comparisons did not include classic noise-robust models, leaving its generalization unverified. Lyu et al. [27] developed an RSG-based soft-threshold denoising model by replacing the BN layers in DRSN-CS with LN layers and substituting FC layers with convolutional layers, demonstrating improved accuracy compared with DRSN-CS and other standard deep-learning networks. Yet, while these advanced models attempt to capture global dependencies, they introduce their own challenges. They typically require vast amounts of training data, possess large parameter counts, and have high computational complexity, making lightweight deployment in resource-constrained industrial settings difficult [28]. Therefore, designing a diagnostic model that is both robust against strong noise and computationally efficient for lightweight deployment remains a critical open problem in intelligent fault diagnosis.

To tackle the aforementioned challenges, this paper presents a novel noise-robust diagnostic network named MWKG-Net, which integrates multi-scale wavelet convolution with kurtosis-guided feature enhancement. This framework is designed to mimic the diagnostic logic of traditional signal processing, thereby fundamentally enhancing the model's feature-extraction capability and robustness in strong-noise environments. Rather than relying on complex network stacking, we introduce two core lightweight modules. First, we develop a multi-scale wavelet convolution block (MWConv1D), which combines the multi-resolution analysis power

of discrete wavelet transforms with the efficiency of depthwise separable convolutions. Within a single module, the network can adaptively decompose the signal into different frequency bands, effectively enlarging its receptive field and capturing multi-scale fault information, while physically separating and suppressing noise. Second, to further accentuate fault-impact features against the noisy background, we design a kurtosis–guided feature re-weighting module (KGFR). This module quantitatively assesses the prominence of fault impacts by computing the spectral kurtosis of the feature maps and dynamically reweights the channels accordingly, guiding the model to focus on the most discriminative impact components. The main contributions of this work can be summarized as follows:

1) An end-to-end lightweight intelligent diagnostic framework called MWKG-Net. Under various noise conditions, especially at extremely low signal-to-noise ratios, this framework achieves significantly higher diagnostic accuracy and robustness compared to a range of state-of-the-art methods.
2) A novel, lightweight multi-scale wavelet convolution module called MWConv1D.By embedding the multi-resolution analysis capability of wavelet transforms into a convolutional network, this module enables efficient multi-scale feature extraction and preliminary noise suppression, substantially enlarging the model's receptive field while reducing computational cost.
3) A kurtosis-guided feature re-weighting module called KGFR. Leveraging kurtosis as a sensitive statistical measure of impact features, this module adaptively amplifies fault-related components and suppresses noise interference. This enhances the model's noise robustness and provides clear physical interpretability.

2. Proposed method

2.1. MWConv1D

To effectively extract multi-scale steady-state and impact features from one-dimensional bearing vibration signals, especially to maintain robustness under strong noise interference, this work is inspired by the denoising approach of Finder et al. [29] and, in combination with bearing fault diagnosis, constructs a novel lightweight one-dimensional wavelet convolution module, MWConv1D. This module leverages the frequency-band decomposition capability of wavelets in combination with depthwise separable convolutions, preserving the physical advantages of wavelet-based frequency division. Through learnable multi-level decomposition, it constructs a multi-scale feature pyramid and uses learnable scaling parameters to automatically regulate the responses of different subband features. Within a single network layer, it achieves multi-scale feature extraction, large receptive-field coverage, and adaptive noise suppression. The model architecture is shown in Fig. 1.

Unlike conventional convolutional networks that rely on stacking multiple layers to enlarge the receptive field, MWConv1D directly builds a multi-scale feature pyramid via cascaded wavelet decompositions. Given a batch of vibration signal samples $X \in \mathbb{R}^{B\times C\times L}$,where B is the batch size, C is the number of channels, and L is the temporal length, this module employs the computationally most efficient Haar wavelet basis for decomposition. Its corresponding low-pass and high-pass decomposition filters can be expressed as:

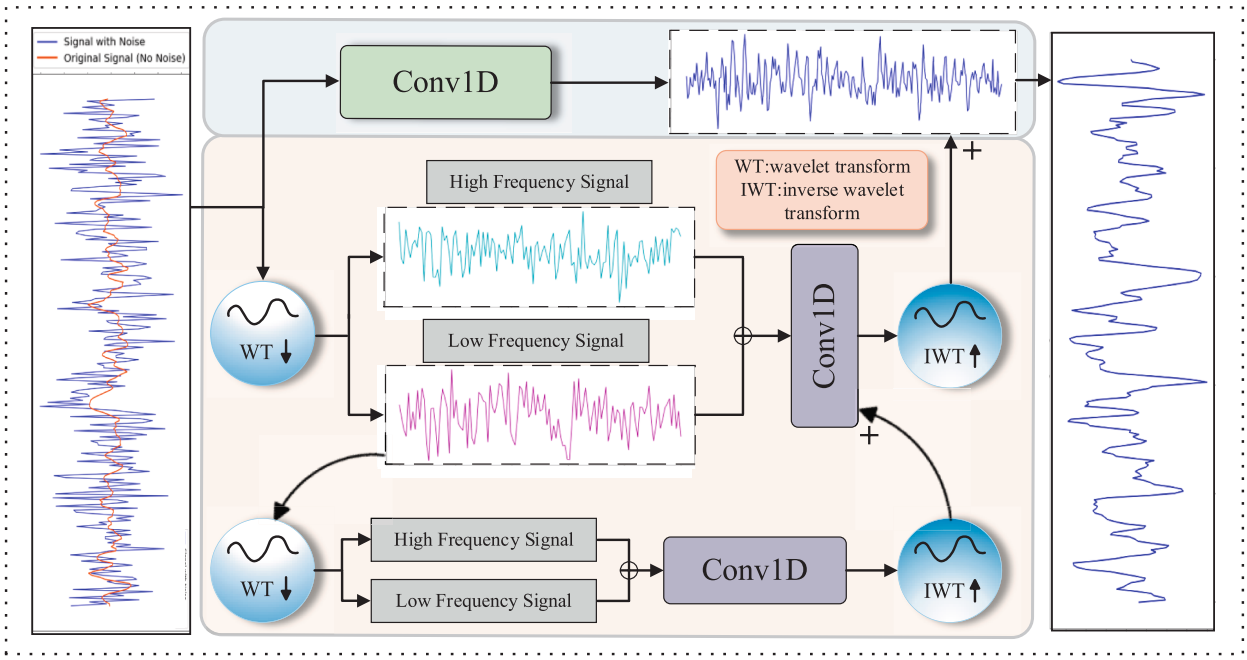


Fig. 1. The structure of MWConv1D.

$$h[n]=\frac{1}{\sqrt{2}}[1,1], g[n]=\frac{1}{\sqrt{2}}[1,-1] \tag{1}$$

For the one-dimensional non-stationary nature of bearing vibration signals, since there is no spatial directionality, we apply a discrete wavelet transform to the input signal X, decomposing it into a low-frequency approximation component X_L and a high-frequency detail component X_H. The approximation X_L preserves the signal's low-frequency steady-state information, such as the ball-pass frequency, inner- and outer-race fault frequencies, and their modulation sidebands, while the detail X_H highlights the signal's high-frequency transient components, such as the non-stationary impact responses generated when rolling elements pass over cracks or spalled regions.

Secondly, to capture fault features at different time scales, MWConv1D performs an L level cascaded decomposition of the low-frequency component X_L:

$$\left[X_{LL}^{(i)}, X_H^{(i)}\right] = WT\left(X_L^{(i-1)}\right), i=1,2,\ldots,L \tag{2}$$

After L levels of decomposition, the signal is organized into a multi-resolution pyramid $\left\{X_H^{(1)}, X_H^{(2)}, \ldots, X_H^{(L)}, X_L^{(L)}\right\}$. At each level i, the low-frequency component $X_L^{(i)}$ and the high-frequency component $X_H^{(i)}$ are concatenated along the channel dimension to form a feature tensor $\overline{X}^{(i)} \in \mathbb{R}^{B\times 2C\times L}$, where $L_i = \lfloor L/2^i \rfloor$. A computationally efficient depthwise separable convolution is then applied to extract features at that scale:

$$z^{(i)} = PWConv\left(DWConv\left(\overline{X}^{(i)}\right)\right) \tag{3}$$

where DWConv [30] is responsible for independently extracting temporal features within each channel, and PWConv, the 1×1 convolution, handles information fusion and interaction across channels. These are shown in Fig. 2. First, the same-scale low-frequency and high-frequency channels are concatenated along the channel dimension. Next, depthwise convolution is applied to extract the temporal features of each individual channel; subsequently, 1×1 convolution is used to perform information fusion across the channels.

On this basis, to enable the network to adaptively handle signals with varying noise levels, a learnable scale modulation module is introduced after the convolution at each scale, with its weight $\alpha^{(i)}$ learned through gradient backpropagation. For the convolution output $z^{(i)}$ at the i scale, the following computation is performed first:

$$\widehat{z}^{(i)} = \alpha^{(i)} \odot z^{(i)}, \alpha^{(i)}\big|_{init} = 0.1 \tag{4}$$

where $\odot$ denotes element-wise multiplication. The learnable weight $\alpha^{(i)} \in \mathbb{R}^{1\times 2C\times 1}$ allows the network to autonomously adjust the importance of low-frequency and high-frequency subbands during training.

After features are independently extracted at each scale, MWConv1D adopts an iterative reconstruction strategy from deep to shallow layers, enabling the fusion of global features captured in deeper layers with local features from shallower ones. Let $r^{(i)}$ denote the intermediate variable during reconstruction, initialized as $r^{(L+1)} = 0$. For the i-th layer (from L in reverse to 1), the reconstruction process is as follows:

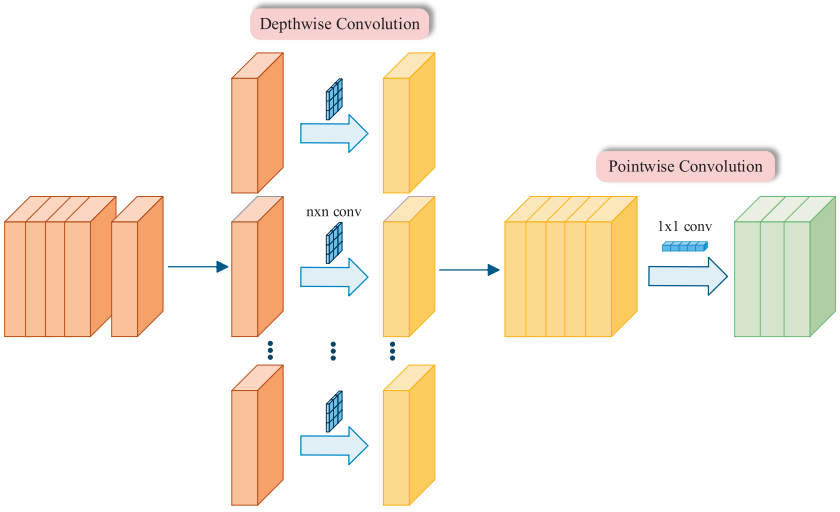


Fig. 2. Depthwise separable convolution.

$$r^{(i)} = IWT\left(\left[\widehat{z}_L^{(i)} + r^{(i+1)}, \widehat{z}_H^{(i)}\right]\right) \tag{5}$$

where $\widehat{z}_L^{(i)}$ and $\widehat{z}_H^{(i)}$ are the low-frequency and high-frequency parts of the modulated feature $\widehat{z}^{(i)}$, respectively. The inverse wavelet transform (IWT) is implemented via transposed convolution as follows:

$$IWT(Y) = ConvTranspose1d(Y, \{g, h\}, stride = 2) \tag{6}$$

The inverse wavelet transform reconstructs the decomposed multi-scale information into feature maps with the same size as the original signal, and transposed convolution is used to enable learnable reconstruction. The final output $r^{(1)}$ is a feature representation that integrates information from all scales and has the same size as the original input.

The MWConv1D module achieves exponential growth in receptive field through L levels of wavelet decomposition. If the kernel size used at each level is k, then after L levels of decomposition, the effective receptive field along the wavelet path is approximately:

$$RF_{eff} \approx k \cdot 2^L \tag{7}$$

Along the layer-wise wavelet paths, the receptive field grows exponentially, which allows covering the modulation cycles of long time windows with only a few layers. This is the core advantage of the multi-scale design. This means that with only a few levels of decomposition, the module can achieve a receptive field far larger than that of conventional large-kernel convolutions, enabling it to cover the full fault modulation cycles in bearing signals, while the number of parameters increases only linearly with L.

The overall computational complexity of the MWConv1D module is primarily determined by the base convolution and the convolutions at each level along the wavelet path. Since all convolutions adopt a depthwise separable structure and the signal length is halved at each decomposition level, the total computational cost can be approximated as:

$$O \approx CKL + \sum_{i=1}^{L} 2CK\frac{L}{2^i} + IWTs \approx O(CL) \tag{8}$$

With the adoption of the depthwise separable structure, the parameters and computational complexity are only linearly correlated with the signal length, and do not increase significantly with the use of large-kernel convolution. It can be seen that the computational cost grows approximately linearly with the signal length L, and is independent of the large effective receptive field. In addition, the learnable parameters of the model mainly come from the depthwise separable convolution kernels at each level and the learnable scaling factors α, without introducing complex attention mechanisms or multi-branch fusion logic. Therefore, the MWConv1D possesses inherent lightweight characteristics, offering low computational cost and high parameter efficiency, while effectively capturing complex feature information.

2.2. Kurtosis-Guided feature re-weighting (KGFR)

Currently, deep learning in bearing fault diagnosis mainly relies on automatically learned feature representations by the network, lacking explicit modeling of fault physical characteristics. This results in poor feature interpretability and insufficient fault discriminability under complex conditions and strong noise environments. To enhance the network's sensitivity to fault impulse features and improve the physical interpretability of features, a Kurtosis-Guided Feature Re-weighting (KGFR) module is proposed, as shown in Fig. 3. Existing feature selection methods lack direct correlation with fault mechanisms. The KGFR module introduces kurtosis, a physical quantity that effectively characterizes signal impulse properties, to quantitatively evaluate the impulse information of feature channels. Based on this evaluation, it dynamically adjusts the weights of each channel, enabling the network to adaptively focus on the impulse features most relevant to faults while suppressing interference from noise and background harmonics.

The KGFR module receives feature maps extracted by the upper network layer as input and outputs a weight-optimized feature map with enhanced fault discriminability through impulse quantification, dynamic feature re-weighting, and dual loss function optimization. For the input feature map $X \in \mathbb{R}^{B\times C\times L}$, the KGFR module first calculates the impulse characteristics of each feature channel x_c.

The standard formula for kurtosis is:

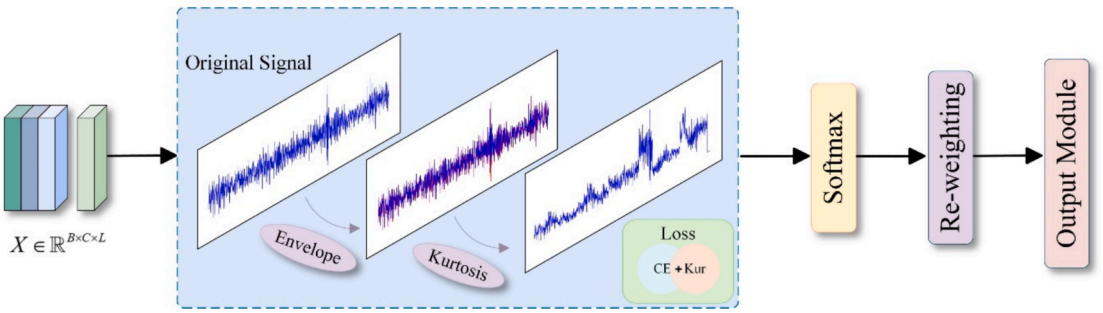


Fig. 3. The structure of KGFR Module.

$$Kurtosis(x_c) = \frac{E\left[(x_c - \mu_c)^4\right]}{\sigma_c^4} \tag{9}$$

where μ_c and σ_c are the mean and standard deviation of channel x_c, respectively.

Since bearing fault features often manifest as periodic modulation of high-frequency resonances, spectral kurtosis is used here as the core metric to more accurately capture impulse information in the envelope. The KGFR module generates a kurtosis vector $K = [k_1, k_2, \ldots, k_C]$ for the input C feature channels. This vector quantitatively describes the amount of impulse information contained in each channel, with higher values corresponding to stronger impulse characteristics.

Based on the kurtosis vector K, the module performs feature re-weighting by generating a set of re-weighting coefficients $\alpha = [\alpha_1, \alpha_2, \ldots, \alpha_C]$ to adjust the importance of each feature channel. These coefficients are derived from the kurtosis vector K through a Softmax [31] function to ensure that all coefficients are positive and sum to 1.

$$\alpha_c = \frac{\exp(k_c)}{\sum_{j=1}^{C} \exp(k_j)} \tag{10}$$

$$\widehat{x}_c = \alpha_c \cdot x_c \tag{11}$$

where x_c represents the re-weighted feature channels.

Finally, all the updated channels together form the new feature map $\widehat{X}$, which is passed to the next network layer. In this way, feature channels with high kurtosis values are assigned greater weights, enhancing their information during subsequent network computations; conversely, channels with low kurtosis values corresponding to noise or background harmonics have their weights suppressed, reducing their influence. This enables the network to adaptively focus its computations on fault-related features.

To maximize the sparsity effect of the KGFR module, the model training process is guided by a composite loss function. This loss function not only includes the standard cross-entropy loss(CE) for supervising the classification task but also introduces a

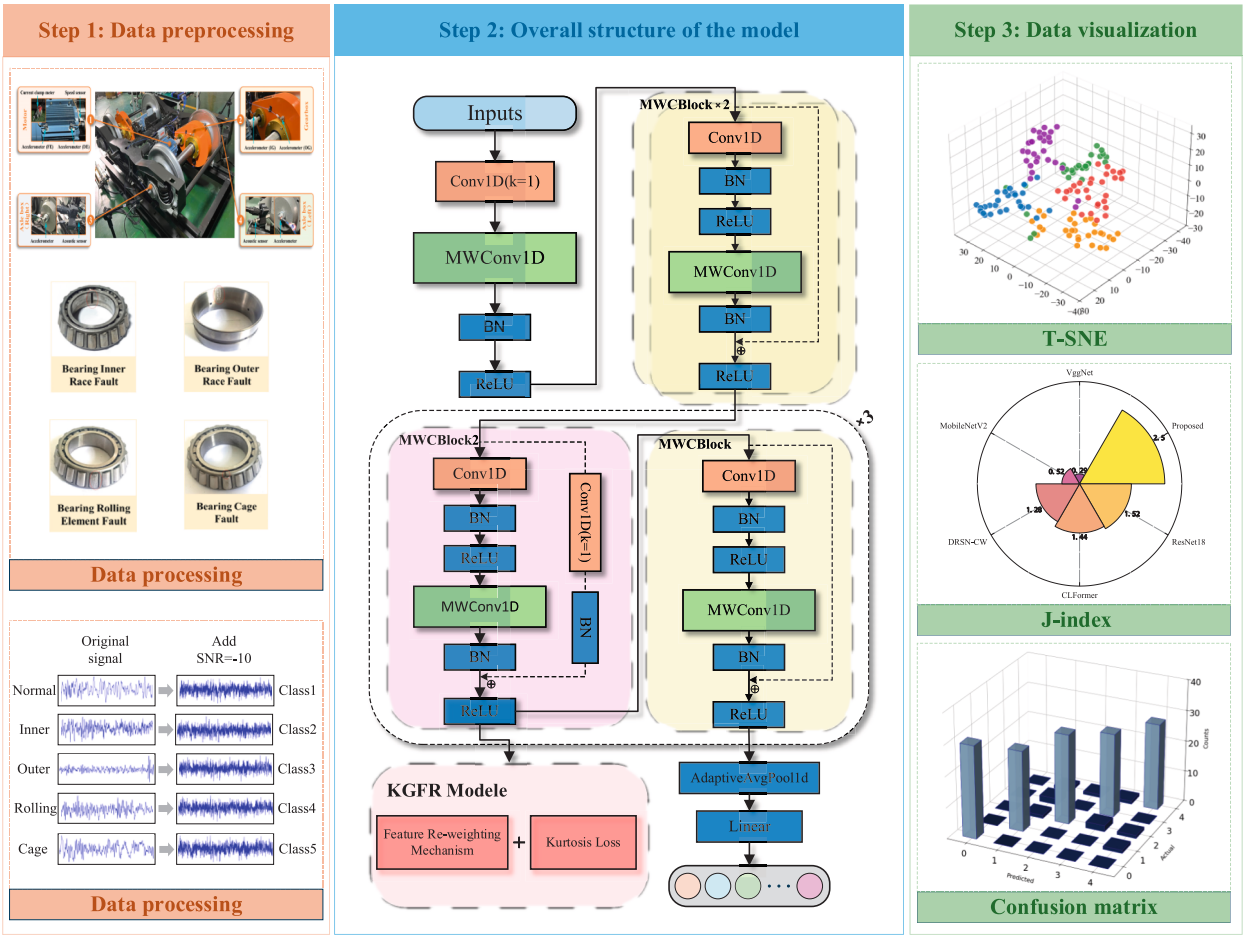


Fig. 4. Overall process of proposed method.

regularization term directly derived from the KGFR module, the kurtosis maximization loss.

The total loss function L_{total} is defined as follows:

$$L_{total} = L_{CE} + \lambda \cdot L_{Kur} \tag{12}$$

where L_{Kur} is defined as the negative average kurtosis of the feature maps output by the module, aiming to maximize the impulsiveness of the features through gradient descent:

$$L_{Kur} = -\frac{1}{C}\sum_{c=1}^{C} Kurtosis(x_c) \tag{13}$$

Moreover, the regularization coefficient λ is not a fixed constant but follows a progressive regularization strategy, where its value is dynamically adjusted over the training epochs:

$$\lambda(e) = \cos\left(\min\left(\frac{e}{E_{total}\cdot\gamma}, 1\right)\cdot\frac{\pi}{2}\right)^2 \tag{14}$$

where e is the current training epoch, E_{total} is the total number of training epochs, and γ is a hyperparameter that controls the decay rate.

In this way, the value of λ is relatively large in the early stages of training, allowing the kurtosis loss to dominate and encouraging the model to prioritize learning and enhancing impulse features in the signal. As training progresses, the value of λ gradually decays to 0, enabling a smooth transition of the optimization objective toward minimizing classification error, and preventing the network from prematurely converging to suboptimal solutions based on secondary features.

2.3. Overall structure

This paper designs a novel end-to-end intelligent fault diagnosis framework, with the overall architecture shown in Fig. 4. The framework mainly consists of the following three steps:

1) Signal preprocessing. Raw one-dimensional vibration signals are collected from an experimental test rig. To enable supervised deep-learning training, a sliding-window technique is applied to the long-duration signals to generate overlapping, fixed-length segments. The resulting dataset is then divided so that 70 % serves as the training set and 30 % serves as the test set for model training and performance evaluation.
2) Overall structure of the model. The preprocessed samples are fed into MWKG-Net. First, they traverse the MWConv1D backbone, which uses built-in wavelet transforms to decompose and reconstruct features at multiple resolutions. This expands the receptive field, captures fault information at various scales, and adaptively suppresses high-frequency noise. The extracted feature maps then enter KGFR, which computes the spectral kurtosis of each channel to quantify fault-related impulses and dynamically reweights channels, enhancing fault features while attenuating noise. During training, the model checkpoint achieving the highest validation accuracy is saved as the final well-trained network.
3) Data visualization. A visual analysis of the results was performed to validate the effectiveness of the proposed model. Fault-diagnosis performance was quantified using accuracy, computational complexity, and the *J*-index. Confusion matrices, T-SNE plots, and bar charts for performance comparison were employed to visually demonstrate the model's superior classification accuracy, lightweight design, and robustness under strong noise and limited-sample conditions.

3. Experimental confirmation

This study utilizes the MFS-MG, BJTU-RAO and Urban Rail Train datasets to evaluate the performance superiority of the proposed model under varying noise conditions. In both datasets, Gaussian white noise with signal-to-noise ratios (SNRs) of −10 dB, −5 dB, 0 dB, 5 dB, and 10 dB was added to the original vibration signals to simulate multi-level noise interference commonly encountered in real-world operating conditions. The choice of additive white Gaussian noise (AWGN) is motivated by two considerations. First, AWGN is a widely used benchmark in signal processing and machine learning, which facilitates direct comparison with previous work. Second, its stationary, broadband statistical properties make it a convenient preliminary testbed for probing noise sensitivity and revealing how the model's performance changes across different noise energy levels. Each fault category contains 100 samples, with 70 samples used for training and the remaining 30 for testing. By evaluating performance under five different noise intensities, the model's robustness, lightweight characteristics, and generalization ability in noisy environments were comprehensively assessed.

Careful hyperparameter selection is essential for effective model optimization and training. In this study we used the Adam optimizer and trained the model for 200 epochs with a batch size of 10 and an initial learning rate of 0.001. All experiments were carried out on a machine equipped with an NVIDIA RTX 4070 GPU.

3.1. Experimental analysis of MFS-MG dataset

3.1.1. Data preparation

The MFS-MG dataset consists of vibration data collected from a mechanical fault comprehensive simulation test platform[32], where the test rig's speed is controlled by a velocity controller. Defective rolling bearings (ER-12) were fabricated using electrical discharge machining. Table 1 lists the corresponding parameters of the rolling bearings. These bearings were installed near the drive motor, and vibration signals were acquired using piezoelectric accelerometers. Fig. 5 shows the test rig used for the MFS-MG dataset and the four types of faulty bearings.

Gaussian white noise with SNRs of −10 dB, −5 dB, 0 dB, 5 dB, and 10 dB was added to the MFS-MG dataset. Each sample has a length of 1024 points, with 70 % of the data used as the training set and the remaining 30 % as the testing set.

During testing, the sampling frequency was set to 12.8 kHz, and the motor rotational speed was maintained at f_r. Based on the basic technical parameters of the bearing and the motor rotational speed, the bearing outer race fault characteristic frequency f_o and the bearing inner race fault characteristic frequency f_i can be calculated using Eqs. (15) and (16), respectively.

$$f_o = \frac{Z}{2}\left(1 - \frac{d}{D}\cos\theta\right)f_r = 3.05f_r \tag{15}$$

$$f_i = \frac{Z}{2}\left(1 + \frac{d}{D}\cos\theta\right)f_r = 4.95f_r \tag{16}$$

In the equations, Z is the number of rolling elements; d is the diameter of each rolling element; D is the pitch diameter; and θ is the contact angle.

3.1.2. Analysis of diagnosis results

The average accuracy and computational complexity of the MFS-MG dataset under different noise levels are shown in Table 2. To validate the effectiveness of the proposed method, we compare it with [33], ResNet18 [34], DRSN-CW [35], MobileNetV2 [36], and VggNet [37]. Among them, CLFormer is a Transformer-based model, while ResNet18, VggNet, and DRSN-CW are classic and stable convolutional neural networks. MobileNetV2 is a representative lightweight convolutional neural network. In the experiments, FLOPs and Params are selected as evaluation metrics for lightweight performance. FLOPs are closely related to the model's computational load and inference time, while Params directly affect storage complexity, training memory usage, and generalization capability. These two indicators assess the model's lightweight nature from the perspectives of time complexity and memory consumption.

Due to the presence of some unknown noise mixed in during the data acquisition process, the quality of the MFS-MG dataset is relatively low. As classic CNN models, ResNet18, DRSN-CW, MobileNetV2, and VggNet can only extract local features and fail to capture sufficient information, resulting in relatively low accuracy. Although CLFormer is capable of extracting global information, its extraction process is hindered by noise embedded in each extraction module, limiting its efficiency in capturing fault feature information. In contrast, the proposed method employs MWConv1D to preliminarily suppress noise and capture multi-scale fault information. Additionally, KGFR dynamically enhances fault impulse features. Using the MFS-MG dataset, the proposed method achieves an accuracy exceeding 92 %, proving to be the best choice across different signal-to-noise ratios and demonstrating strong robustness.

T-SNE provides an intuitive visualization of fault diagnosis results [38]. This dimensionality-reduction technique preserves the local relationships of high-dimensional data, so that similar samples cluster closely in the low-dimensional space while dissimilar ones are pushed apart. In fault classification, tight clustering of same-class points alongside clear separation of different-class points indicates superior performance.

Fig. 6 presents the 3D T-SNE visualization results of various models trained on the MFS-MG dataset. The proposed method shows tight intra-class clustering and clear inter-class separation. The samples within each class are closely grouped, and the distances between classes are noticeably larger. Although a few misclassified samples remain, the overall clustering quality is high, allowing effective differentiation of different fault data. In contrast, other models display overlapping class distributions with blurred boundaries. Some fault types are hard to distinguish or even unrecognizable. For instance, models like CLFormer and ResNet18 suffer from significant inter-class overlap. This further confirms that the proposed method outperforms others in feature extraction capability.

To highlight the lightweight nature of the proposed method, Fig. 7 displays a bubble chart comparing the accuracy and computational complexity of various models on the MFS-MG dataset. In this chart, the vertical position of each bubble indicates the parameter count or FLOPs, while the bubble size reflects the model's accuracy. Based on this visualization, the following conclusions can be drawn:

1) Among the eight models, our method achieves the highest accuracy while using fewer parameters and lower FLOPs.

Table 1
The parameters of ER-12.

Inner ring diameter/mm	Outer ring diameter/mm	Rolling element diameter /mm	pitch circle diameter/mm	Number of rolling element	Contact Angle/(°)
25.4	52	7.9375	33.4772	8	0

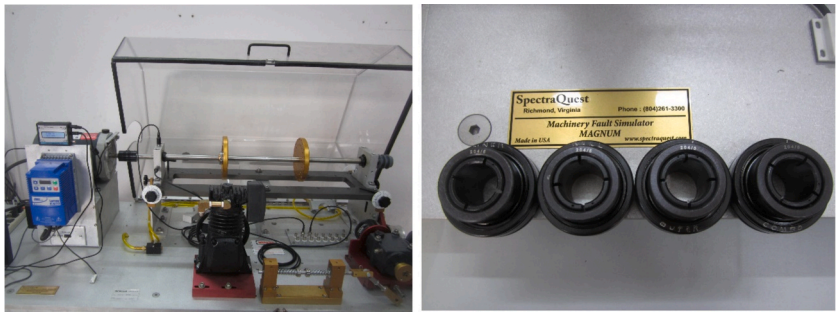


Fig. 5. MFS-MG test bench and faulty bearing.

Table 2
Average accuracy and computational complexity of different methods on MFS-MG dataset.

Models	SNR/dB					Complexity	
	−10	−5	0	5	10	FLOPs(M)	Params(M)
Proposed	92.67	94.00	94.67	97.46	98.00	83.01	1.82
CLFormer	86.00	89.33	90.67	92.67	93.33	4.76	0.21
ResNet18	79.33	86.67	88.67	90.00	93.33	175.68	3.85
DRSN-CW	76.27	86.00	91.33	94.00	95.67	177.22	5.25
MobileNetV2	68.79	77.33	86.00	90.67	93.33	111.00	2.18
VggNet	56.53	67.33	68.00	77.33	78.27	789.27	11.93

2) Although models such as ResNet18 and DRSN-CW deliver similar accuracy and diagnostic performance, they require substantially more parameters and FLOPs, resulting in increased computational overhead. This comparison underscores the lightweight advantages of our approach on noisy datasets.

3.2. Experimental analysis of BJTU-RAO dataset

3.2.1. Data preparation

The BJTU-RAO dataset, developed by Beijing Jiaotong University, is the world's first publicly available dataset simulating faults in rail transit train bogie transmission systems [39]. In this experiment, vibration signals from the left axle box under the WC2 condition (motor speed 40 Hz, lateral load 0 kN) were analyzed. Fig. 8 shows the equipment setup of the BJTU-RAO test rig. The axle box bearing model is HRB 352213, with a sampling frequency set at 64 kHz. The experimental data used consists of five fault categories, as detailed in Table 3, with bearing fault types classified into five classes based on fault location.

3.2.2. Analysis of diagnosis results

On the BJTU-RAO dataset, noise testing of the model was conducted using the same noise addition method as for the MFS-MG dataset. The average accuracy and computational complexity under different noise levels are shown in Table 4. Visualization of the experimental results is presented in Fig. 9, which displays bar and line charts of diagnostic accuracy and computational complexity, respectively, for various methods under −10 dB, −5 dB, and 0 dB noise levels on the BJTU-RAO dataset. Models are ordered from left to right by increasing computational complexity; the bars depict diagnostic accuracy, and the complexity curves are plotted on a nonlinear scale for clearer visualization. From these results, we can conclude:

1) With respect to accuracy, the proposed method achieves superior learning effectiveness and fault classification performance. Even under strong noise conditions of −10 dB, it maintains a high fault recognition accuracy, consistently above 90 %. In contrast, other models exhibit lower accuracy under −10 dB noise due to the masking of fault features by the high noise intensity.
2) From the perspective of computational complexity, the proposed method has fewer parameters and lower FLOPs than the lightweight model MobileNetV2. Our method achieves higher accuracy than ResNet18, with 3.8 million parameters and 175 million FLOPs, clearly underscoring its lightweight advantage.

Next, confusion matrices are plotted for intuitive analysis of classification results, as shown in Fig. 10. The "Predicted" axis denotes the model's predicted labels, and the "Actual" axis shows the ground-truth labels. Taller bars along the diagonal correspond to more accurate predictions and superior classification performance. The figure illustrates that our method achieves notably higher diagonal bars than ResNet18, whose shorter diagonal bars and increased off-diagonal counts reveal poorer predictive accuracy.

The results under the SNR = −10 dB condition are visualized using ROC curves in Fig. 11. The ROC curve of the proposed model is closest to the top-left corner, indicating its superior performance compared to other models under high noise conditions. The False

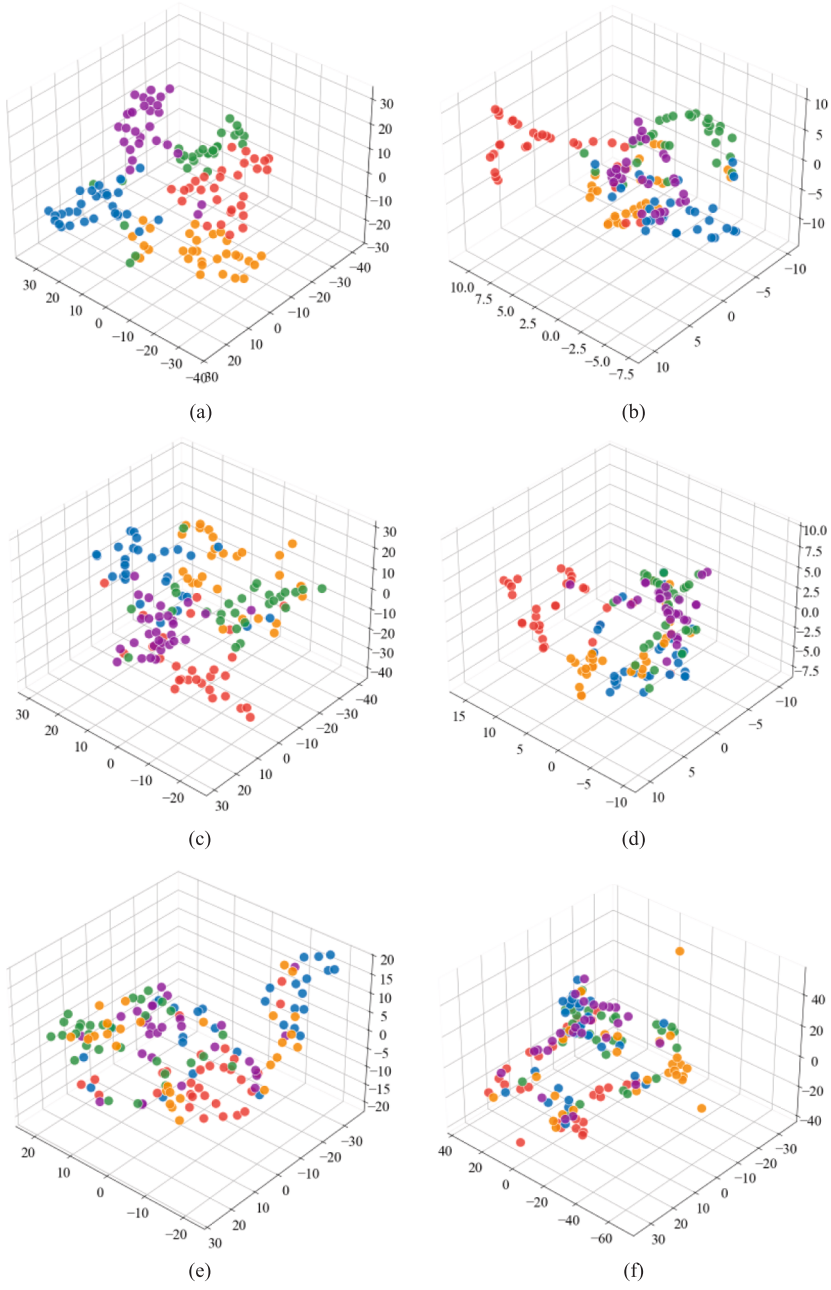


Fig. 6. MFS-MG dataset TSNE three-dimensional visualization results. (a) Proposed; (b) CLFormer; (c) ResNet18; (d) DRSN-CW; (e) MobileNetV2; (f) VggNet.

Positive Rate (FPR) and True Positive Rate (TPR), which correspond to the horizontal and vertical axes of the ROC curve respectively, are calculated as follows:

$$TPR = \frac{TP}{TP + FN} \tag{17}$$

$$FPR = \frac{FP}{FP + TN} \tag{18}$$

The subsequent evaluation uses the J-index [40], which is expressed as follows:

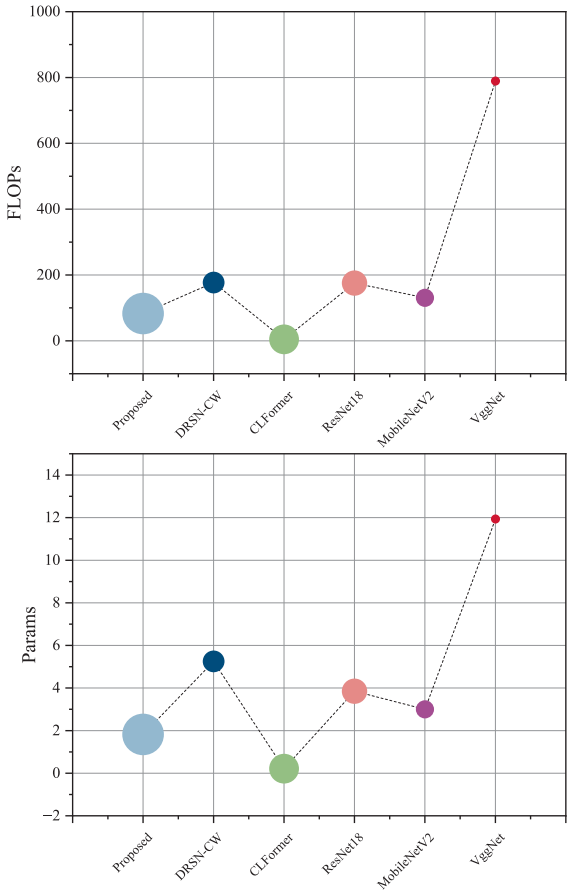


Fig. 7. Bubble chart of accuracy and computational complexity for different methods on the MFS-MG dataset.

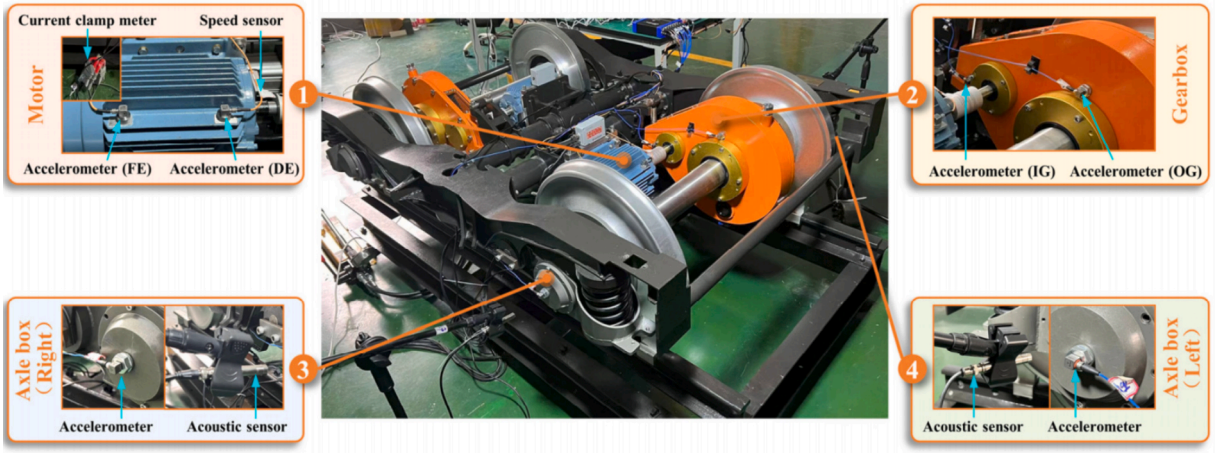


Fig. 8. Experimental platform of subway train bogies.

$$J = \frac{\|S_b\|}{\|S_w\|} \tag{19}$$

$$S_w = \sum_{c=1}^{C} \sum_{i=1}^{N_c} \left(f_i^c - \bar{f}^c\right)\left(f_i^c - \bar{f}^c\right)^T \tag{20}$$

Table 3
Classes of BJTU-RAO fault.

Bearing condition	Label
Normal condition	0
Bearing inner race fault	1
Bearing outer race fault	2
Bearing rolling element fault	3
Bearing cage fault	4

Table 4
Average accuracy and computational complexity of different methods on BJTU-RAO dataset.

Models	SNR/dB					Complexity	
	−10	−5	0	5	10	FLOPs(M)	Params(M)
Proposed	93.33	97.33	100	100	100	83.01	1.82
CLFormer	84.67	94.67	98.26	98.67	100	4.76	0.21
ResNet18	82.00	92.27	96.27	98.00	99.33	175.68	3.85
DRSN-CW	78.67	89.60	96.40	98.67	99.00	177.22	5.25
MobileNetV2	69.33	83.20	94.93	95.33	99.33	111.00	2.18
VggNet	52.67	59.47	62.80	80.67	82.67	789.27	11.93

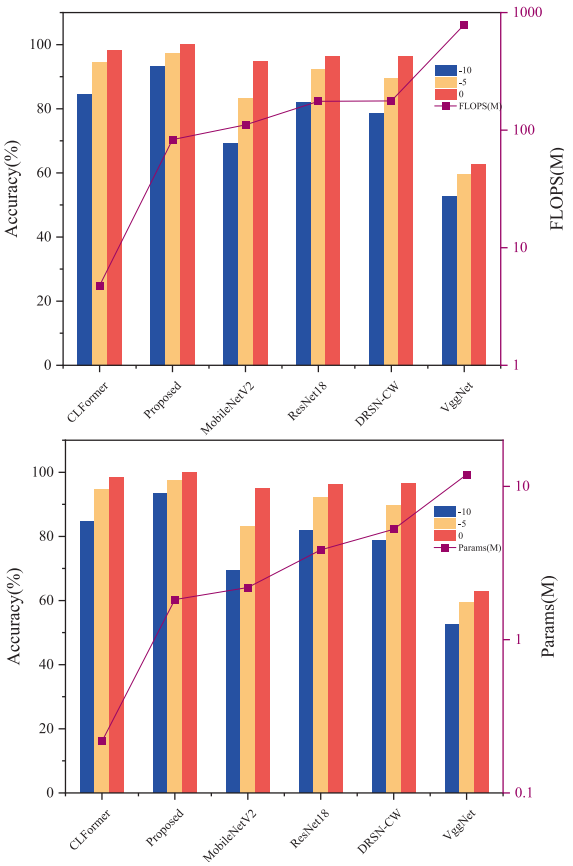


Fig. 9. Diagnostic accuracy and computational complexity of different methods.

$$S_b = \sum_{c=1}^{C} N_c (\bar{f}^c - \bar{f})(\bar{f}^c - \bar{f})^T \tag{21}$$

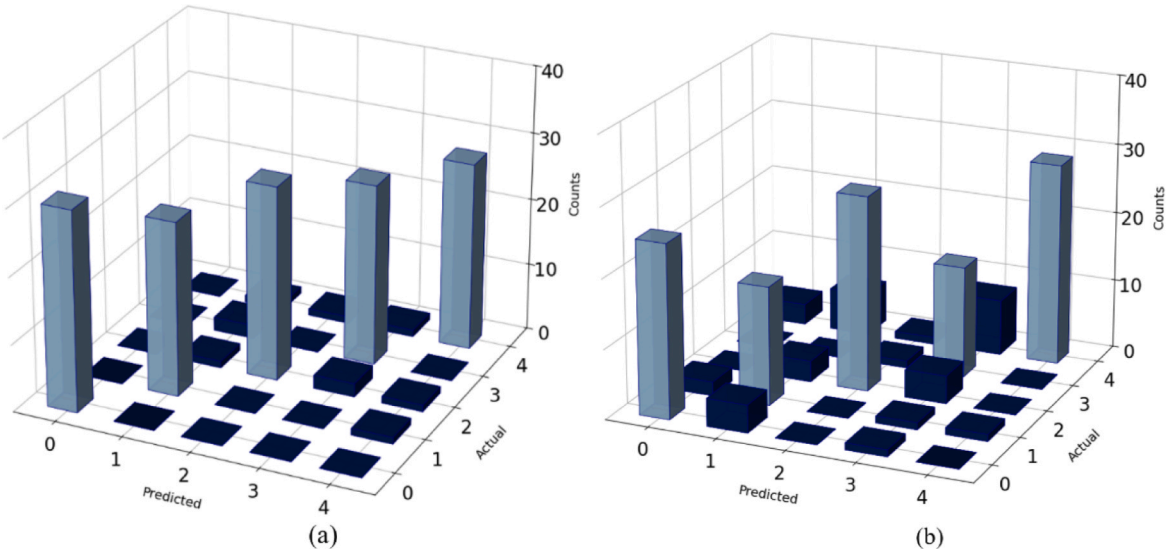


Fig. 10. Confusion matrix of model predictions after training on the BJTU-RAO dataset. (a) Proposed method; (b) ResNet18.

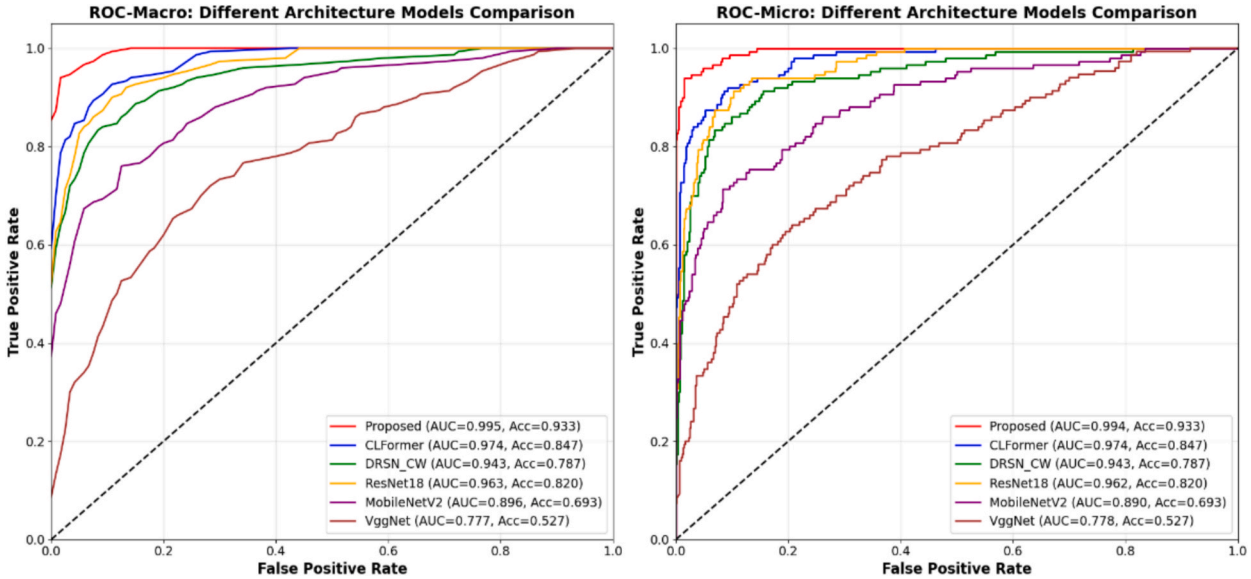


Fig. 11. ROC curves of different models on the BJTU-RAO dataset.

$$\bar{f}^{c} = \frac{1}{N_c}\sum_{i=1}^{N_c} f_i^{c} \tag{22}$$

Table 5
J-index of different models across BJTU-RAO dataset.

Method	BJTU-RAO	
	mean	min
Proposed	2.50	2.17
CLFormer	1.44	1.12
ResNet18	1.52	0.98
DRSN-CW	1.28	0.82
MobileNetV2	0.52	0.23
VggNet	0.29	0.20

$$\bar{f} = \frac{1}{N}\sum_{i=1}^{N} f_i \tag{23}$$

This equation incorporates the within-class covariance S_w, the between-class covariance S_b, and a second-order normalization term. In this context, C denotes the total number of classes, N the total number of samples, f_i the output feature of the i-th sample, and $\bar{f}$ the mean feature vector across all samples. The J-index quantifies the clustering behavior of the data. A higher J-index indicates well-separated clusters, where samples from the same class are tightly grouped and those from different classes are clearly distinct, reflecting superior classification performance. In contrast, a lower J-index implies poor separability, with overlapping classes and scattered intra-class points. Unlike accuracy, the J-index evaluates classification quality from the perspective of cluster structure.

Table 5 lists the J-index of different models on the BJTU-RAO dataset. It can be observed that the proposed method achieves the highest J-index among all models, highlighting its superior classification performance under high-noise conditions.

Fig. 12 displays the results in a radar chart, where the radius of each sector reflects the J-index of each method, with a longer radius indicating a higher J-index. Fig. 13 presents the outcomes of five repeated trials using violin plots, with black dots marking the results of individual experiments. As shown, the J-index of the proposed method consistently ranges between 2 and 3, while most other methods fall between 0.5 and 2, highlighting their weaker performance. A closer look reveals that the proposed method not only achieves a generally high J-index but also shows minimal variation across trials, suggesting stable and reliable clustering. In contrast, methods like CLFormer and ResNet18 have lower and more scattered J-index, indicating limited clustering ability when handling the data. These results emphasize the superior classification performance of the proposed method.

We performed a systematic ablation study to validate the effectiveness of the proposed MWConv1D and KGFR modules. To trace the sources of the performance gains, we removed these modules from MWKG-Net and compared the models at noise levels of -10, -5 and 0 dB. The accuracy curves are shown in the Fig. 14.

Experimental results show the proposed modules are highly complementary for noise-robust feature extraction. The wavelet-based multi-scale convolution separates and characterizes time–frequency features at different scales, improving modeling of multi-band and short transient events. The kurtosis-based KGFR module adaptively recalibrates channel importance using statistical cues, emphasizing impact components that carry fault information while suppressing noise. When fused in sequence, the two modules effectively enrich multi-dimensional and fault-specific features, and the model achieves more robust discrimination and higher recognition rates in various low-SNR tests.

3.3. Experimental analysis of urban rail train dataset

3.3.1. Data preparation

To realistically evaluate and improve the model's generalization, we introduced a rolling-bearing dataset collected from an urban rail train in China. The data were acquired on an operational line and naturally include continuous variations in speed, load, and operating stage. During collection, the train ran on the depot track at 79 km/h. The sampling frequency was set to 10 kHz. The dataset contains five bearing conditions: healthy, inner-ring fault, outer-ring fault, rolling-element fault, and cage fault. Table 6 lists the bearing parameters, and Fig. 15 shows the rolling bearing from the urban rail train.

3.3.2. Analysis of diagnosis results

On this dataset, we tested the model under noise using the same noise-addition procedure applied to the two datasets above. The mean accuracies and computational complexity under different noise levels are shown in Table 7. To comprehensively evaluate MWKG-Net, we compared it with MWA-CNN [41] and MCKD-DeNet [42]. MWA-CNN is an advanced fault-diagnosis model that uses a discrete wavelet transform layer to extract useful features, which are then fed into a convolutional neural network (CNN) for further feature learning. MCKD-DeNet adopts maximum-correlated kurtosis as its loss function to adaptively enhance and extract periodic impact fault features from highly noisy signals. The excellent performance of the model can be seen more intuitively from the radargram in Fig. 16.

From this figure, models exhibit notable performance variations under different noise levels on the Urban Rail Train Dataset. The proposed model shows obvious accuracy advantages across multiple signal-to-noise ratio (SNR) conditions, delivering even more prominent performance especially in high-noise scenarios (SNR $= -10$ dB). While MWA-CNN and MCKD-DeNet maintain a certain level of accuracy under some noise levels, MWA-CNN relies on discrete wavelet transform for feature extraction and lacks generalization capability when dealing with signals containing multiple fault types and noise in complex scenarios. MCKD-DeNet uses maximum correlated kurtosis loss to enhance periodic impact features, yet it fails to cover diverse fault types. Owing to its more comprehensive design in feature learning and noise robustness, the proposed model can accurately capture features of multiple fault types and adapt to different SNRs, resulting in superior overall accuracy. Additionally, other models such as CLFormer and ResNet18 all present gaps to varying degrees compared with our model in terms of accuracy or noise robustness.

To assess the effectiveness of the kurtosis regularization term, we designed a controlled experiment. We compared two models with the same network architecture: one trained with the standard cross-entropy loss and the other with the proposed regularization term added. Both models used the same optimizer, learning-rate schedule, and number of training epochs. We trained and evaluated them on the same dataset to verify the impact of the proposed regularizer on model performance. The results are shown in Table 8.

Present the results as a stacked bar chart, as shown in Fig. 17. It can be seen that the CE + Kur bars are longer than those of CE-only at all noise levels, indicating that adding the kurtosis regularizer improves accuracy across SNRs. The gain is especially pronounced at

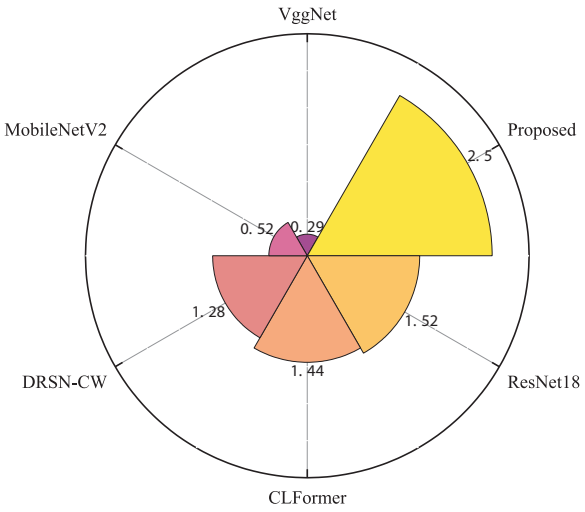


Fig. 12. Rose diagrams of J-index for different models on the BJTU-RAO dataset.

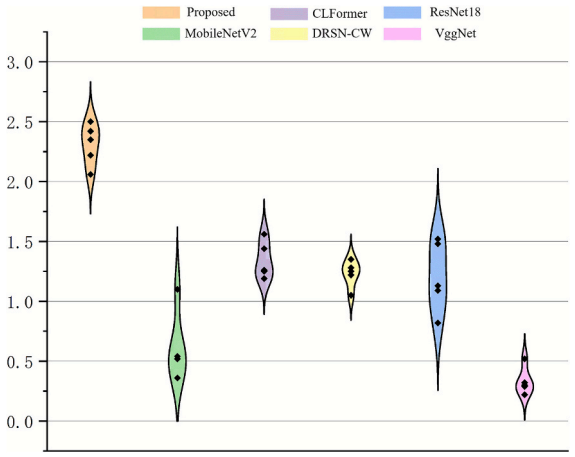


Fig. 13. Violin plot of J-index for different models on the BJTU-RAO dataset.

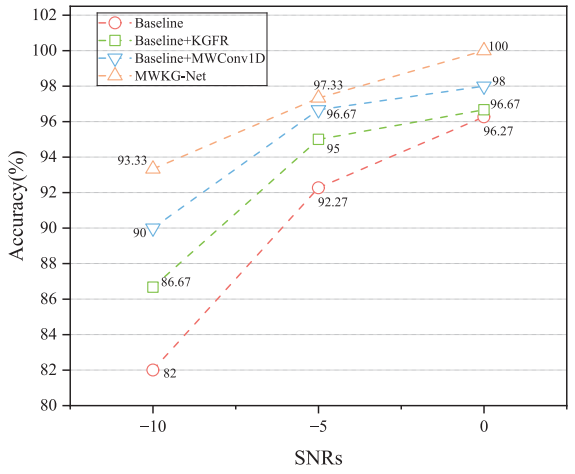


Fig. 14. Ablation experiment accuracy line chart.

Table 6
Bearing parameters.

Model	Inner diameter (mm)	Outer diameter (mm)	Rolling-element diameter (mm)	Pitch diameter (mm)	Number of rolling elements
BC1B 326,441 A/HB1	130 mm	220 mm	26 mm	176 mm	18



Fig. 15. Rolling bearing of the Urban Rail Train.

Table 7
Average accuracy and computational complexity of different methods on Urban Rail Train Dataset.

Models	SNR/dB					Complexity	
	−10	−5	0	5	10	FLOPs(M)	Params(M)
Proposed	85.33	89.33	92.00	95.67	97.00	83.01	1.82
CLFormer	69.33	75.33	84.76	87.00	92.33	4.76	0.21
ResNet18	62.00	74.33	84.67	86.67	90.33	175.68	3.85
DRSN-CW	58.00	74.00	80.33	83.33	88.67	177.22	5.25
MobileNetV2	66.67	69.55	75.00	76.33	80.00	111.00	2.18
VggNet	25.33	48.00	50.33	53.00	55.33	789.27	11.93
MWA-CNN	64.00	79.67	87.00	89.67	91.76	96.59	1.98
MCKD-DeNet	61.00	77.00	88.33	89.33	91.67	196.3	6.99

low SNR (−10 dB).

3.4. Deployment performance evaluation

To validate the lightweight characteristics and real-time inference capability of the proposed model, we conducted measurements on an NVIDIA GeForce RTX 4070. We evaluated latency, throughput, peak GPU memory usage, and model size. During testing, the model was set to eval() mode and executed under torch.no_grad(). Each experiment used 50 warm-up runs and 200 timed runs, with an input size of 1×1024.

From Table 9, it can be seen that the proposed model achieves a significant reduction in parameter count and inference latency while maintaining high diagnostic accuracy. With a batch size of 1, the model reaches an average la'tency of only 4.541 ms and a throughput of 223.22 samples/s. Its model size is just 3.15 MB, approximately one-fourteenth that of standard ResNet18. These results demonstrate the strong lightweight characteristics and practical deployment potential of the proposed model.

4. Discussion

While MWKG-Net has achieved encouraging results in experiments, there is still room for improvement in its performance boundaries and intrinsic characteristics.

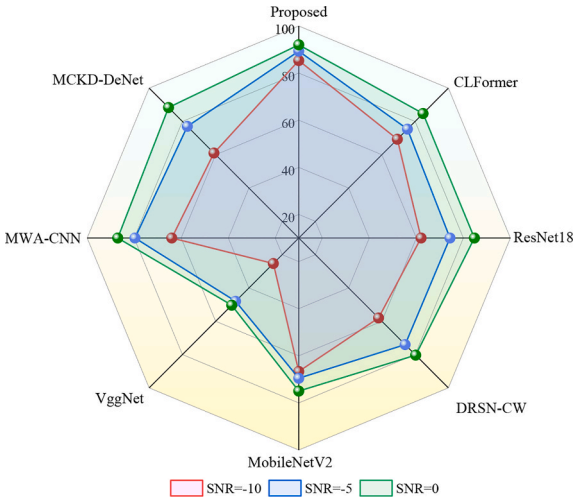


Fig. 16. Radar chart of accuracy for different methods on the Urban Rail Train Dataset.

Table 8
Average accuracy of different regularization terms on the Urban Rail Train Dataset.

Model	SNR/dB		
	−10	−5	0
Ce + kur	85.33	89.33	92.00
Ce-only	79.00	84.33	87.67

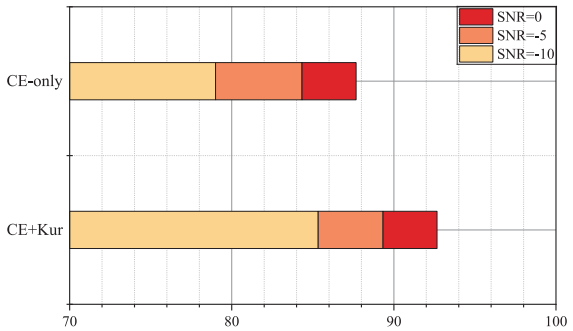


Fig. 17. Stacked bar chart of accuracy for different regularization terms on the Urban Rail Train Dataset.

Table 9
Comparison of lightweight and real-time inference metrics between the proposed model and ResNet18 on NVIDIA GeForce RTX 4070.

Model	Batch	Latency (ms)	Throughput (samples/s)	Peak Mem (MB)	Model Size (MB)	Params (M)
Proposed	1	4.541 ± 0.90	223.22	12.7	3.15	1.82
ResNet18	1	21.4 ± 1.2	46.7	110.3	44.8	11.7

First, the model shows limited robustness under extremely low signal-to-noise ratio (SNR) conditions. Although its diagnostic accuracy at below −10 dB remains notably higher than that of comparative models, it decreases compared to its performance at higher SNR levels. This indicates that when noise intensity far exceeds the energy of fault impacts, the current feature enhancement mechanism struggles to handle the situation. To address this issue, future work will focus on enhancing robustness in extremely low-SNR environments. Efforts will be made to develop more powerful noise estimation and suppression strategies, improving the model's ability to discriminate features under severe noise.

Second, the model's performance in real embedded environments has yet to be practically verified. While its modular structure contributes to favorable lightweight characteristics, actual power consumption and inference speed on highly resource-constrained embedded devices still require validation through real deployment tests. In response to this limitation, follow-up research will

promote deeper industrial application and deployment. The model will be implemented on actual industrial embedded platforms, where long-term monitoring will assess its diagnostic accuracy, inference latency, and power consumption in real operating environments. This will help complete the transition from algorithmic innovation to industrial application, while also exploring its potential in diagnosing compound faults.

5. Conclusion

To tackle the difficulty of detecting weak fault features and the low recognition rates of traditional models in noisy industrial environments, this paper introduces MWKG-Net—a lightweight and noise-robust diagnostic network for rolling bearings. This end-to-end framework integrates signal processing principles with deep learning, allowing direct identification of fault types from raw vibration signals without complex manual feature design.

Extensive tests on the MFS-MG, BJTU-RAO, and Urban Rail Train datasets confirm that MWConv1D effectively mimics the multi-resolution analysis of wavelet transforms. It enlarges the receptive field, captures multi-scale fault information, and suppresses noise interference. KGFR uses kurtosis to guide the model toward impactful impulse components, significantly improving feature robustness. Compared to leading models such as CLFormer, ResNet18, DRSN-CW, and MobileNetV2, MWKG-Net consistently excels across all evaluation metrics. Even under extreme noise conditions (SNR $= -10$ dB), it still maintains the highest diagnostic accuracy in our experiments, while using significantly fewer parameters and lower computational cost (FLOPs), highlighting its noise robustness, efficiency, and lightweight design.

MWKG-Net not only surpasses mainstream convolutional networks and advanced Transformer models but also provides an efficient and reliable solution for accurate fault diagnosis under high-noise environments. Given its lightweight architecture, future work will involve further research into its compatibility and adaptability for embedded hardware platforms. This exploration is a crucial step toward assessing its potential for real-time online monitoring in industrial settings.

CRediT authorship contribution statement

Yurui Jia: Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Dechen Yao:** Writing – review & editing, Funding acquisition. **Jianwei Yang:** Writing – review & editing. **Limin Jia:** Writing – review & editing. **Tao Zhou:** Supervision. **Quanyu Long:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Notation of key symbols and terms used in the proposed method.

Table
Notation of key symbols and terms used in the proposed method.

Notation	Description
X	Input batch of vibration signals
$h[n]$	Low-pass decomposition filter
$g[n]$	High-pass decomposition filter
B	Batch size
C	Number of input channels
L	Temporal length of the signal
i	Number of decomposition layers
$\alpha^{(i)}$	Learnable scaling weight at the i-th layer
$z^{(i)}$	Convolution output at the i-th level
$r^{(i)}$	Intermediate variable during reconstruction at the i-th laye
K	Vector of spectral kurtosis values
μ_c	Mean of the channel
σ_c	Variance of the channel

(*continued on next page*)

Table (*continued*)

Notation	Description
λ	Regularization coefficient in the loss function
e	Current training epoch
$\odot$	Element-wise multiplication

Data availability

The authors do not have permission to share data.

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